



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO KOWRAPPA GOUNDER & SONS FUEL AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

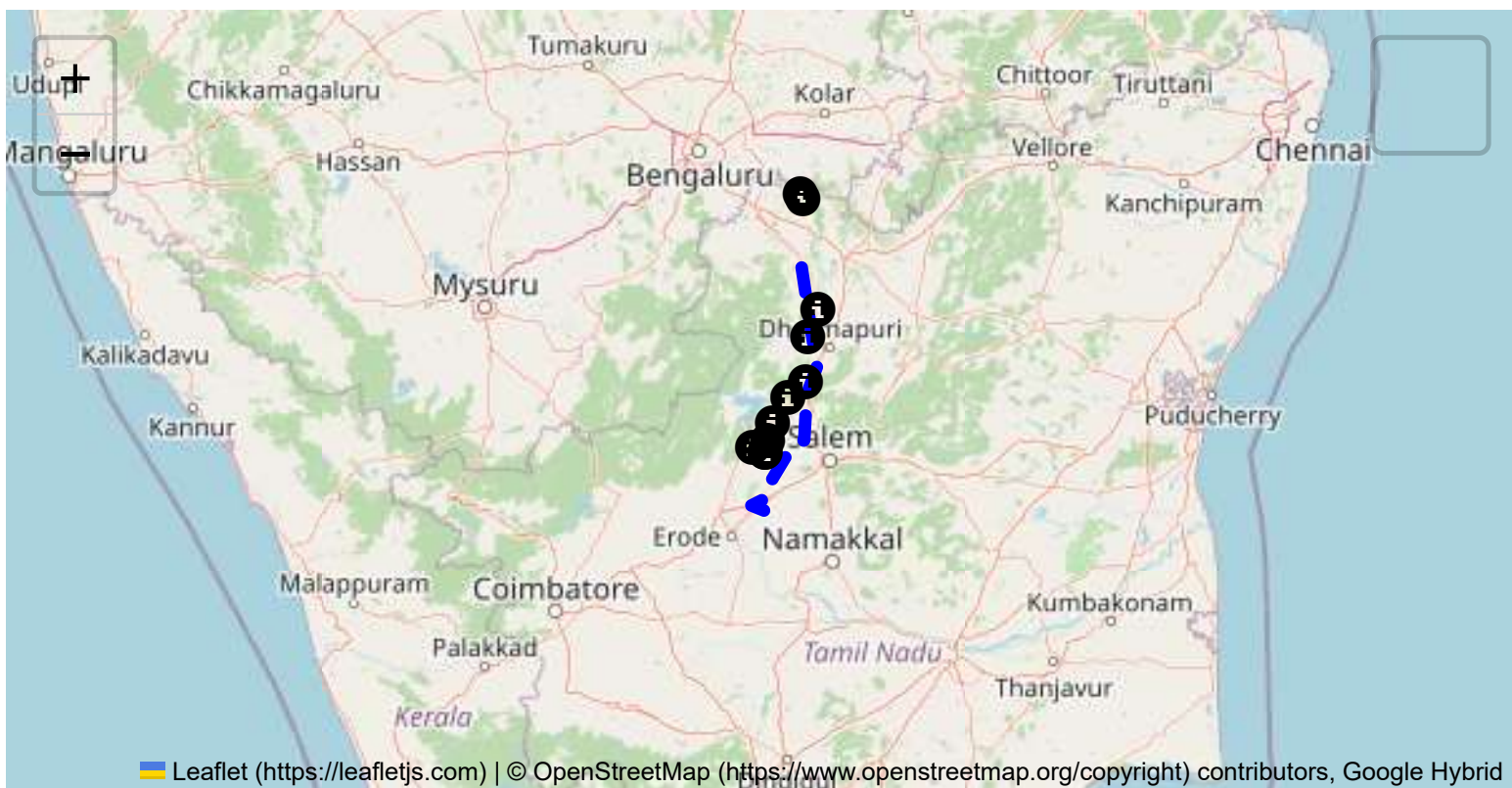
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

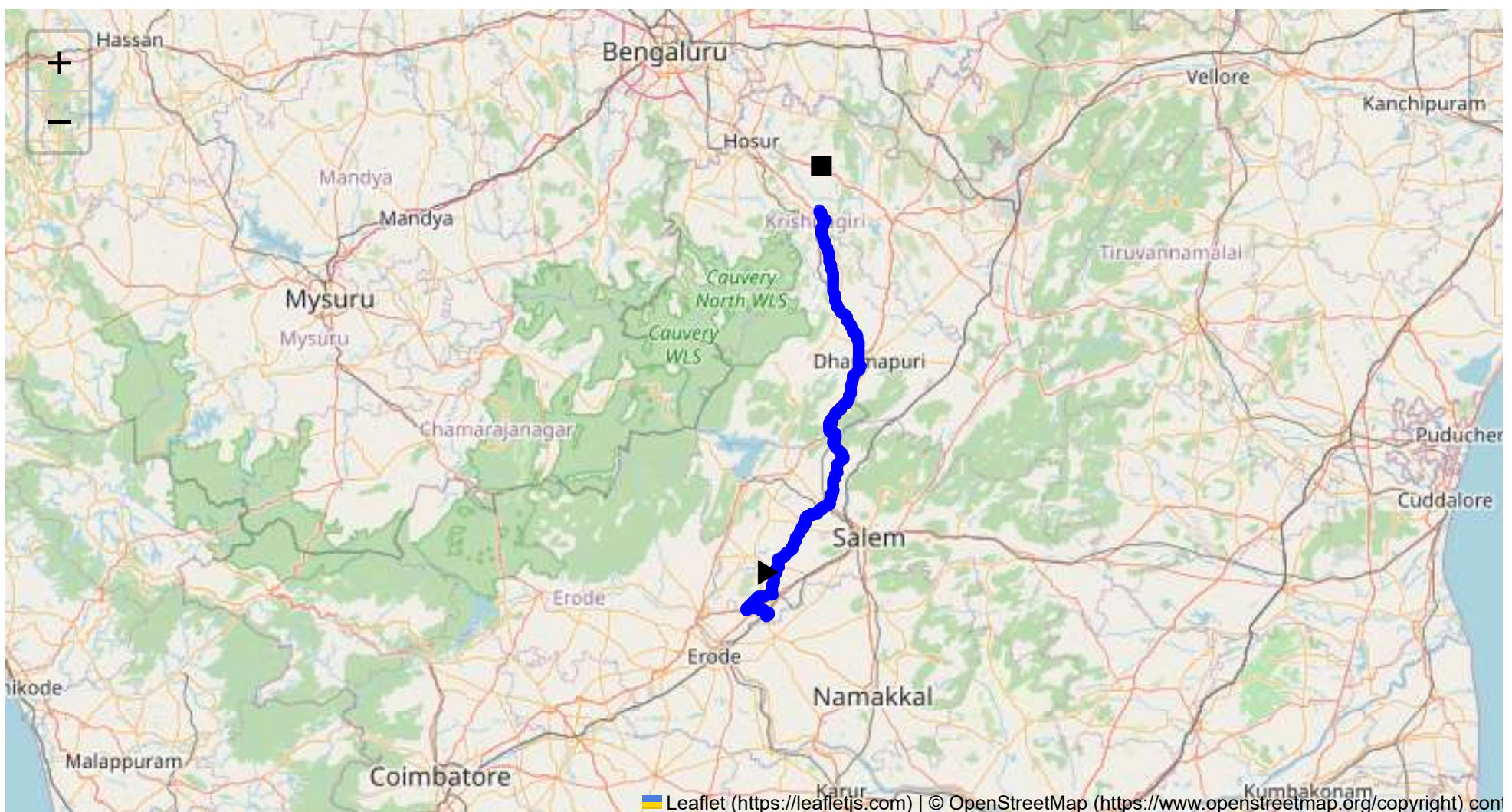
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 153.18 km
Estimated Duration: 3.0 hours
Adjusted Duration (Heavy Vehicle): 3.8 hours
Start: (11.4381, 77.8734)
End: (12.542723, 78.020975)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Mugalur spans roughly 153.18 kilometers and is expected to take about 3.05 hours for heavy vehicles carrying hazardous materials. It passes through several towns and rural areas, using both state highways and local roads, which may include various terrains such as flatlands and gentle slopes typical of Tamil Nadu's landscape.

Weather Conditions

- **Typical Weather:** Tamil Nadu experiences a tropical climate. The region can be hot and dry from March to May, with monsoon rains from June to September.
- **Potential Hazards:** During the monsoon, there may be waterlogging and reduced visibility which can affect driving conditions. In summer months, heat may cause road surface issues and fatigue.

Traffic Patterns

- **Peak Hours:** Traffic is usually heavier during morning (8-10 AM) and evening (5-8 PM) rush hours, particularly near Salem and other urban areas along the route.
- **Congestion-prone Areas:** Congestion is likely in and around Salem and Omalur due to local traffic and bottlenecks at junctions.

Road Quality and Infrastructure

- **Road Conditions:** The Salem - Kochi Highway and Omalur-Sankagiri Road are generally well-paved, though local roads near villages might be narrow with varied maintenance levels.
- **Infrastructure:** Be mindful of local roads having fewer amenities like petrol stations or rest areas, particularly as you proceed away from major towns.

Alternative Routes

- In case of emergencies or roadblocks, consider using the NH44 from Salem to Dharmapuri, which is a National Highway and likely better maintained and faster.

Local Regulations for Hazardous Materials

- Transport of hazardous materials in Tamil Nadu requires adherence to specific routes and timings as prescribed by local transport authorities. Ensure compliance with local traffic rules and permits for hazardous materials.

Historical Incidents

- The Salem region has occasional reports of accidents involving heavy vehicles due to human error, vehicle failure, or poor weather. There's a low incidence of hazardous material-specific incidents.

Environmental Considerations

- Be cautious of areas near waterbodies and wildlife sanctuaries such as reserved forest areas adjacent to roads. Noise and pollution should be minimized to reduce ecological impact.

Communication Coverage

- **Coverage Assessment:** Major highways like those through Salem have good mobile network coverage. Rural stretches might experience patches of weak signal, and measures should be taken for

satellite communication tools in emergencies.

Emergency Response Times

- Response times could vary, with urban areas like Salem offering quicker support (typically under 30 minutes) compared to remote village areas which might experience 45 minutes to an hour or more delay.

Risk Assessment Summary

Overall, this route poses moderate risk primarily due to traffic congestion and potential weather-related challenges. Awareness of traffic patterns, compliance with regulations, and preparedness for varying road conditions are crucial to ensure safe passage. Regular updates on weather and traffic are recommended. Emergency protocols should be rehearsed with a focus on areas with limited communication.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	Medium	11.45543, 77.81740	30 KM/Hr
7	Turn	High	11.45584, 77.81695	15 KM/Hr
8	Turn	Medium	11.45656, 77.81709	30 KM/Hr
9	Turn	High	11.49383, 77.88442	15 KM/Hr
10	Turn	Medium	11.56083, 77.89921	30 KM/Hr
11	Turn	Medium	11.56180, 77.90019	30 KM/Hr
12	Turn	Medium	11.59189, 77.90645	30 KM/Hr
13	Turn	Medium	11.59328, 77.90809	30 KM/Hr
14	Turn	Medium	11.63867, 77.94841	30 KM/Hr
15	Turn	High	11.64217, 77.95200	15 KM/Hr
16	Turn	Medium	11.64712, 77.95118	30 KM/Hr
17	Turn	Medium	11.67858, 77.96834	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
18	Turn	High	11.70233, 77.97869	15 KM/Hr
19	Turn	Medium	11.71394, 77.99572	30 KM/Hr
20	Turn	Medium	11.71916, 78.01117	30 KM/Hr
21	Turn	Medium	11.72504, 78.01680	30 KM/Hr
22	Turn	Medium	11.73766, 78.03445	30 KM/Hr
23	Turn	High	11.73651, 78.03868	15 KM/Hr
24	Turn	Medium	11.74257, 78.04708	30 KM/Hr
25	Turn	Medium	11.77998, 78.05786	30 KM/Hr
26	Turn	Medium	11.86915, 78.08382	30 KM/Hr
27	Turn	Medium	11.87998, 78.08170	30 KM/Hr
28	Turn	Medium	11.91266, 78.05852	30 KM/Hr
29	Turn	Medium	11.93746, 78.05331	30 KM/Hr
30	Turn	Medium	11.95531, 78.04698	30 KM/Hr
31	Turn	High	11.95854, 78.04990	15 KM/Hr
32	Turn	High	11.96015, 78.04945	15 KM/Hr
33	Turn	Medium	11.96370, 78.05142	30 KM/Hr
34	Turn	Medium	11.96433, 78.05113	30 KM/Hr
35	Turn	High	11.96542, 78.04938	15 KM/Hr
36	Turn	High	11.96650, 78.04927	15 KM/Hr
37	Turn	Medium	12.12027, 78.13284	30 KM/Hr
38	Turn	Medium	12.50427, 78.03115	30 KM/Hr
39	Turn	Medium	12.51389, 78.03229	30 KM/Hr
40	Turn	Medium	12.51548, 78.03394	30 KM/Hr
41	Turn	High	12.51751, 78.03410	15 KM/Hr
42	Turn	High	12.51773, 78.03256	15 KM/Hr
43	Turn	High	12.51966, 78.03042	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	hospital	Government Hospital, Jakkasamudram	12.3937246, 78.0522641	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45584, 77.81695



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45656, 77.81709



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56083, 77.89921



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56180, 77.90019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59189, 77.90645



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59328, 77.90809



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70233, 77.97869



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71394, 77.99572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71916, 78.01117



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72504, 78.01680



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73766, 78.03445



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03868



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74257, 78.04708



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.77998, 78.05786



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86915, 78.08382



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87998, 78.08170



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.91266, 78.05852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96015, 78.04945

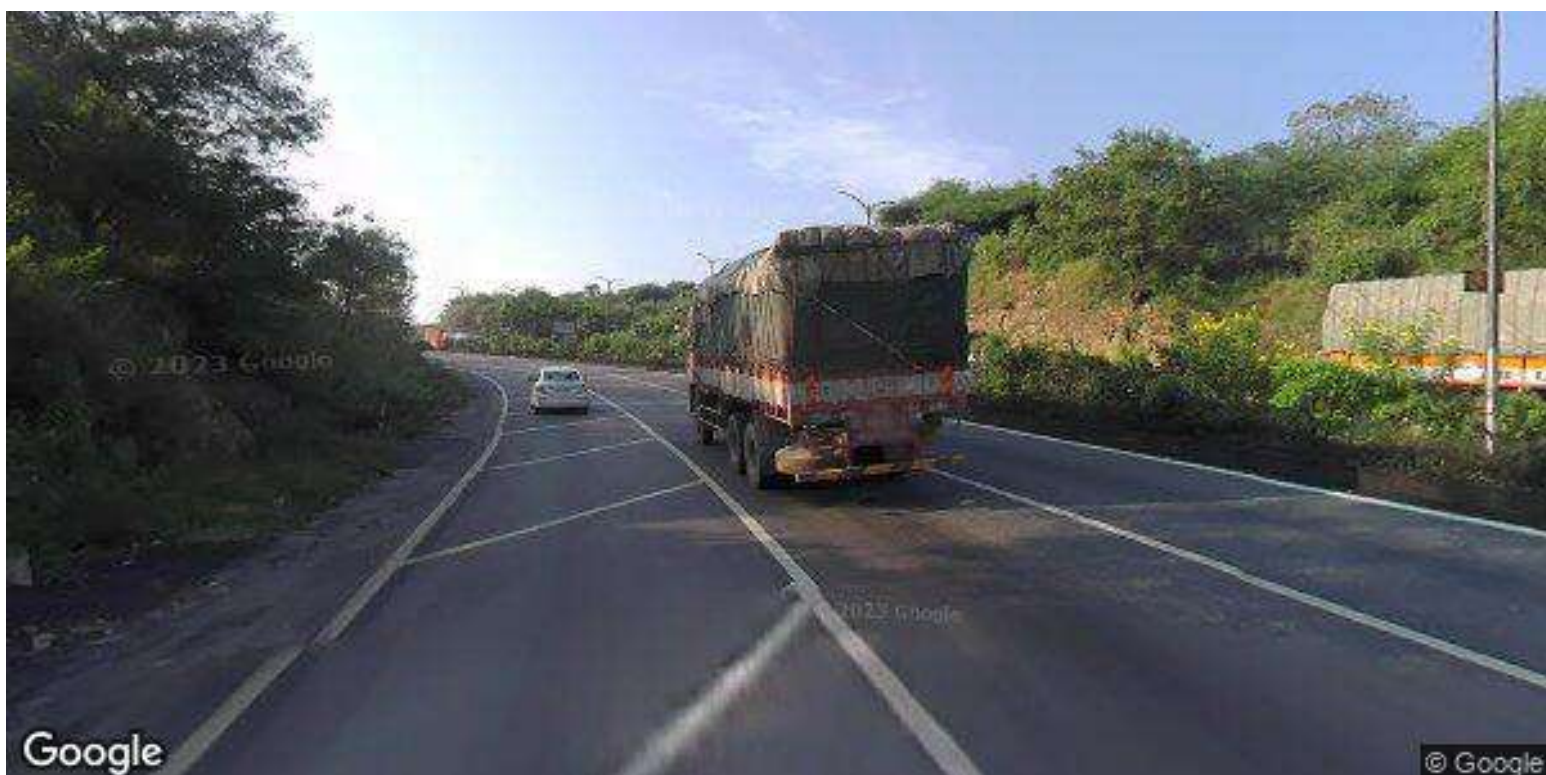


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96370, 78.05142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.12027, 78.13284



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.50427, 78.03115



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51389, 78.03229



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51548, 78.03394



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.51751, 78.03410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.51773, 78.03256



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.51966, 78.03042

